

SOP-11: Reproductive Toxicants and Bioactive Small Molecules (PHS)

*Roberts Lab, School of Aquatic and Fishery Sciences - Rev. 0 - September 1, 2026***PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES** - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 1 - Lab-Specific Information

Building/Room(s) covered by this SOP	Fisheries Teaching & Research (FTR) Building, Room 209
Unit or department	School of Aquatic and Fishery Sciences (SAFS), College of the Environment
Principal Investigator	Steven Roberts (work phone 206-685-3742)
PI signature / date	_____ Date: _____
This SOP was created by (if not PI)	_____ Date: _____
Emergency contact	9-1-1 (campus phone) EH&S 206-543-7262 (M-F 8-5) 206-685-UWPD (8973) after hours
Date of last review	_____ Next review due: _____ (review annually and after any incident or process change)

Chemicals covered by this SOP

Chemical name (as listed in MyChem)	CAS number
4-Methoxyphenol (mequinol)	150-76-5
Serotonin creatinine sulfate monohydrate	61-47-2
Acetylcholine chloride	60-31-1
Resazurin sodium salt (two containers, 15 g and 50 g)	62758-13-8
Poly(I:C) (HMW), TLR3 agonist	not classified / proprietary

Safety Data Sheets for every chemical listed above must be readily available in the laboratory. SDSs are accessible through MyChem.

Section 2 - Hazards

Applicable GHS pictograms: Health hazard (reproductive toxicity Cat 2); Exclamation mark (irritant, sensitizer, acute toxicity Cat 4); Environment (aquatic toxicity)

- PARTICULARLY HAZARDOUS SUBSTANCE. 4-Methoxyphenol and serotonin creatinine sulfate are both classified GHS Reproductive toxicity Category 2 (suspected of damaging fertility or the unborn child). Reproductive toxins are one of the three categories that define a particularly hazardous substance. Sections 9-11 of this SOP are mandatory and apply to all compounds covered here.

- Stock chemicals: 4-methoxyphenol (100 g solid), serotonin creatinine sulfate monohydrate (1 g solid), acetylcholine chloride (25 g solid), resazurin sodium salt (15 g and 50 g solids), poly(I:C) HMW (50 mg). Intermediates and working solutions: aqueous or DMSO stock solutions and dilute working solutions in filtered seawater or buffer, used to induce spawning in bivalves (serotonin), to test cholinergic responses (acetylcholine), as a metabolic viability indicator (resazurin/AlamarBlue), as an immune stimulant (poly(I:C)), and as a polymerization inhibitor or test compound (4-methoxyphenol). Wastes: unused solids, stock and working solutions, treated seawater and assay plates, and contaminated debris.
- The common hazard across this group is powder handling. All five are supplied as solids or lyophilized powders, and weighing them generates respirable dust of compounds that are reproductive toxicants, sensitizers or biologically active at low doses.
- Routes of exposure: inhalation of dust (primary), skin contact and absorption, eye contact, ingestion.
- How exposure might occur: weighing powder, transferring solid, dissolving and vortexing, pipetting stock solutions, dosing assay plates or spawning containers, and contact with contaminated balance or bench surfaces.
- Target organs: 4-methoxyphenol - reproductive system, skin (sensitization), eyes, and the aquatic environment (acute aquatic toxicity Category 2, chronic Category 3). Serotonin creatinine sulfate - reproductive system; it is a neurotransmitter and is biologically active at low doses. Acetylcholine chloride - eyes, skin, respiratory tract; as a cholinergic agonist it can in principle produce systemic cholinergic effects at high dose. Resazurin sodium salt - eyes, skin, respiratory tract (irritant). Poly(I:C) - a synthetic double-stranded RNA immune stimulant; it is a potent interferon inducer and can cause a febrile inflammatory response if injected or absorbed, and is treated as biologically active rather than chemically toxic.
- Signs and symptoms of exposure: eye redness and irritation; skin redness, itching or allergic dermatitis (4-methoxyphenol is a skin sensitizer Category 1); coughing and airway irritation from dust (resazurin and acetylcholine chloride are STOT-single Category 3, respiratory irritation). Reproductive effects produce no acute signs.
- GHS classification (from MyChem / SDS): 4-Methoxyphenol - Acute toxicity Category 4 (oral); Reproductive toxicity Category 2; Skin sensitization Category 1; Serious eye damage/eye irritation Category 2A; Acute aquatic toxicity Category 2; Chronic aquatic toxicity Category 3. Serotonin creatinine sulfate - Reproductive toxicity Category 2. Acetylcholine chloride - Serious eye damage/eye irritation Category 2A; Skin corrosion/irritation Category 2; STOT-single Category 3 (respiratory irritation). Resazurin sodium salt - Serious eye damage/eye irritation Category 2A; Skin corrosion/irritation Category 2; STOT-single Category 3 (respiratory irritation). Poly(I:C) - not classified.
- No OSHA PEL, WISHA PEL or ACGIH TLV has been established for any of these compounds. Control is by dust containment, designated area and work practice.

Section 3 - Engineering Controls and Personal Protective Equipment (PPE)

Engineering controls

- ALL weighing and transfer of these solids, and all preparation of stock solutions, is performed inside the certified chemical fume hood in FTR 209 with the analytical balance placed inside the hood, on a disposable absorbent liner, with the sash at or below the marked working height. Any chemical fume hood used must be tested and passed by EH&S. Weighing these powders on an open bench balance is prohibited.
- Where the supplier offers the compound pre-weighed or as a solution, that form is preferred and avoids the powder-handling step entirely.
- Dissolution and vortexing are done with the tube capped, inside the hood.
- Once in solution, dilute working solutions may be handled at the bench or in the biological safety cabinet for cell and larval work.
- Poly(I:C) is handled with aseptic technique in the biological safety cabinet after reconstitution, both to protect the reagent and to control aerosol.

- The nearest eyewash and safety shower are located in FTR 209 and are inspected and flushed weekly by lab staff.

Hygiene measures

- Avoid contact with skin, eyes, and clothing. Wash hands after removing PPE, before breaks, and immediately after handling the chemical. If any chemical covered by this SOP comes into contact with any PPE, the PPE shall be immediately removed and discarded properly. Any potentially exposed body parts should be washed immediately.
- No eating, drinking, chewing gum, applying cosmetics, or storing food or drink in any area where these chemicals are used or stored.

Skin and body protection

- Chemically compatible laboratory coats that fully extend to the wrist must be worn, appropriately sized and buttoned to their full length. Personnel must also wear full-length pants, or equivalent, and close-toe shoes. The area of skin between the shoe and ankle must not be exposed.
- Lab coat type: buttoned barrier lab coat. Because 4-methoxyphenol is a skin sensitizer and a suspected reproductive toxicant, disposable chemically resistant sleeves are worn for all powder weighing, and a chemically resistant apron plus face shield is worn for spill cleanup.
- Any lab coat visibly contaminated with these powders is removed, bagged and laundered through the SAFS service or discarded; contaminated coats are never taken home.

Hand protection

- Hand protection IS required for the activities described in this SOP.
- Required gloves: DOUBLE 4-6 mil nitrile gloves for weighing and handling 4-methoxyphenol and serotonin creatinine sulfate powders, because both are reproductive toxicants and 4-methoxyphenol is a sensitizer. Single 6-8 mil nitrile gloves are acceptable for acetylcholine chloride, resazurin and poly(I:C) powders and for all dilute working solutions.
- Consult your preferred glove manufacturer to ensure that the gloves you plan to use are compatible with the specific chemical being used. Gloves must be inspected prior to use, including a check for pinholes.
- Change gloves immediately on any visible contact with powder or solution, and before leaving the designated area.
- Use proper glove removal technique (without touching the glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices. Wash and dry hands immediately after glove removal.

Eye protection

- ANSI Z87.1-compliant eye protection IS required for all work with these chemicals. Ordinary prescription glasses will NOT provide adequate protection unless they also meet the Z87.1 standard and have compliant side shields.
- Minimum eye protection: ANSI Z87.1 chemical splash goggles for all powder weighing and stock preparation. Safety glasses with side shields are acceptable for handling dilute working solutions at the bench. A face shield over goggles is required for spill cleanup.

Respiratory protection

- Respiratory protection is NOT required for the activities described in this SOP, provided all powder handling is confined to the certified chemical fume hood as written.
- These compounds in POWDER form are never permitted to be handled outside of the chemical fume hood. Dilute working solutions may be used at the bench or in the biological safety cabinet.
- Respiratory protection may be needed if a dust or aerosol hazard is present and work is conducted outside of the fume hood; because such work is prohibited here, no respirator is needed. If any procedure may pose such a hazard, it should be eliminated or strictly isolated.

- Respirators should be used as a last line of defense (i.e., after engineering and administrative controls have been exhausted), and when any Action Limit (AL) or Occupational Exposure Limit (OEL) has been exceeded or when there is a possibility that an AL/OEL will be exceeded. If a potential exposure hazard cannot be eliminated, contact the EH&S Respiratory Protection Program administrator to discuss respiratory protection or to enroll in the program so a respiratory protection analysis can be performed. Program enrollment includes medical evaluation, training and fit testing for an appropriate respirator.
- Where air-purifying respirators are appropriate, use a full-face respirator with a P100 particulate filter as a backup to engineering controls.

Section 4 - Special Handling and Storage Requirements

- Quantity limits: no more than 5 g of any of these solids is weighed in a single operation, and stock solutions are prepared at the smallest practical volume.
- Prepare single-use aliquots of stock solutions so that the powder container is opened as few times as possible over the life of the reagent.
- 4-Methoxyphenol is a skin sensitizer: any lab member who has previously reacted to it must not handle it, and the PI must be informed so the task can be reassigned.
- 4-Methoxyphenol is acutely toxic to aquatic life (Category 2). Never discharge it, or seawater containing it, to a sink, a drain, a seawater return line, or any system connected to natural waters. This is the single most important environmental control in this SOP.
- Serotonin creatinine sulfate used for spawning induction is prepared and injected or applied with care; used spawning seawater is collected as chemical waste and is not returned to a flow-through system.
- All containers - powder bottles, stock tubes, working solution bottles and waste containers - must be labeled with the full chemical name, concentration, solvent, the applicable hazard words ('REPRODUCTIVE TOXIN', 'SENSITIZER', 'TOXIC TO AQUATIC LIFE'), the date and the preparer's initials.
- Powder weighing is performed only when at least one other person is present in or immediately adjacent to FTR 209. Unattended operations with open powder are prohibited; assay plates and treated cultures incubating in closed, labeled vessels are permitted.
- A 'REPRODUCTIVE TOXICANT IN USE - PARTICULARLY HAZARDOUS SUBSTANCE' sign is posted on the fume hood sash while 4-methoxyphenol or serotonin is being weighed.
- Storage: solids are stored in labeled, closed secondary containers - 4-methoxyphenol in the toxics cabinet in FTR 209 at room temperature, protected from light; serotonin creatinine sulfate, acetylcholine chloride and poly(I:C) refrigerated or frozen per the manufacturer; resazurin at room temperature protected from light. Secondary containers holding reproductive toxicants are posted with the PHS designation. Segregate from oxidizers and strong acids.
- Transport: containers travel in closed, leak-proof secondary containment.
- Clean the chemical fume hood upon completion of tasks: remove and bag the absorbent liner, then wipe the work surface and balance with a damp wipe followed by detergent solution, clean water and 70% ethanol; wipe dry.
- Clean all contaminated surfaces with detergent solution followed by clean water and dry. Do not brush or blow powder residue.
- Place all contaminated disposable items in appropriate laboratory waste for disposal, in a closed container.
- Non-disposable/re-usable utensils, glassware, and other surfaces contaminated with these compounds must be decontaminated at the end of the laboratory work session. Complete this inside the chemical fume hood before removing any of the items.
- When work is completed, remove gloves and wash hands with soap and water.

Section 5 - Spill and Accident Procedures

Spill response

- Chemical spills must be cleaned up as soon as possible by properly protected and trained personnel. All other persons should leave the area. Do not attempt to clean up any spill if not trained or comfortable. Evacuate the area and call 9-1-1 on a campus phone for help.
- POWDER spill: do NOT brush, sweep or wipe dry, which aerosolizes the material. If the spill is contained inside the operating fume hood and is less than approximately 5 g, trained staff may clean it up: dampen with water using a wash bottle, wipe up with damp absorbent working from the outside inward, and place all material in a screw-top waste container. Wash the surface with detergent solution followed by clean water.
- A powder spill outside the fume hood, or larger than approximately 5 g: restrict access, post a sign, and contact EH&S at 206-543-0467 during business hours or via 9-1-1 after hours.
- LIQUID spill - less than 100 mL of a dilute working solution: may be handled by trained staff wearing goggles, nitrile gloves and a lab coat. Absorb with absorbent pads from the outside inward, place in a screw-top waste container, and wash the area with detergent solution and clean water.
- Any spill of a concentrated stock solution, any liquid spill greater than 100 mL, and any spill in which a person may have been exposed: restrict access, post a sign, and contact EH&S.
- Any spill of 4-methoxyphenol or a solution containing it that reaches or could reach a sink, floor drain, or seawater line must be reported to EH&S immediately because of its acute aquatic toxicity, even if the quantity is small.
- PPE required for cleanup: chemical splash goggles, face shield for powder or concentrated stock, double nitrile gloves, buttoned lab coat with disposable chemically resistant sleeves and apron, full-length pants and close-toe shoes.
- Signage and entry restrictions: post 'DO NOT ENTER - CHEMICAL SPILL' on the door for any powder spill outside the hood; restrict entry until EH&S clears the space.
- Spill area must be cleaned up in the following manner: after absorbent removal, clean the area thoroughly with detergent solution followed by clean water, then 70% ethanol, and dry.
- Spill cleanup materials must be disposed of in the following manner: double bag all absorbent and contaminated disposables, place in a screw-top container, label with a UW Hazardous Waste Label naming the compound and 'spill debris', and submit a collection request to EH&S.
- For questions on spill cleanup, contact EH&S spill consultants at 206-543-0467 during normal business hours (Monday-Friday, 8 a.m. to 5 p.m.).
- Any spill, exposure or near miss incident requires the involved person or supervisor to complete and submit an incident report on the EH&S website within 24 hours (certain types of incidents require immediate notification).

Exposure procedures - first aid

- Perform first aid immediately. Refer to the SDS for additional chemical-specific guidance.
- Skin exposure: remove contaminated gloves and clothing and wash the affected skin with soap and copious running water for at least 15 minutes, using the safety shower for anything more than a small contact; use a clean lab coat or spare clothing for cover-up. Seek medical evaluation for any rash or persistent irritation, particularly after 4-methoxyphenol contact.
- Eye exposure: use the eyewash for 15 minutes while holding the eyelids open; seek medical evaluation.
- Inhalation exposure: move the person out of the contaminated area to fresh air; get medical help if coughing or irritation persists.
- Ingestion: rinse mouth with water; do not induce vomiting; call 9-1-1.

- Sharps injury or accidental injection: this is significant for serotonin, acetylcholine and poly(I:C), all of which are biologically active. Scrub the exposed area thoroughly for 15 minutes using warm water and sudsing soap and seek immediate medical evaluation; call 9-1-1 if any systemic symptoms develop.
- Get help. Call 9-1-1 or go to the nearest Emergency Department; provide details of exposure: agent, dose, route of exposure, and time since exposure. Bring the SDS and this SOP to the Emergency Department.
- Notify your supervisor as soon as possible for assistance. Secure the area before leaving; lock doors and indicate the spill if needed.
- Notify EH&S immediately after providing first aid and/or getting help. During business hours (M-F, 8-5) call 206-543-7262. Outside of business hours call 206-685-UWPD (8973) to be routed to EH&S staff on call.

Section 6 - Waste Accumulation and Disposal Procedures

- Waste streams: (a) unused solids; (b) concentrated stock solutions; (c) dilute working solutions, treated seawater and spent assay plates; (d) contaminated solid debris (weigh boats, spatulas, tips, plates, gloves, liners).
- 4-Methoxyphenol and any solution containing it must NEVER be poured down the drain or discharged to a seawater system - it is acutely toxic to aquatic life at Category 2. Collect all of it as hazardous waste.
- Seawater used for serotonin-induced spawning, and any seawater containing these compounds, is collected as chemical waste rather than returned to a flow-through or recirculating system, unless EH&S has confirmed in writing that a specific dilute solution may be sewerred.
- Spent resazurin/alarBlue assay plates and dilute acetylcholine solutions may contain no other hazardous constituent; nonetheless, collect them as chemical waste unless EH&S guidance for that specific solution permits sewer disposal.
- Accumulate liquid waste at the point of generation in a sturdy, chemically compatible HDPE container with a securely closable screw-top lid, kept closed except when actively adding waste, in secondary containment.
- All chemical waste containers must be labeled with a UW Hazardous Waste Label listing every constituent and approximate percentage, starting from the moment the first drop of waste is added. Refer to 'How to Label Chemical Waste Containers' on the EH&S website.
- Solid debris is collected in a lined, closed, labeled solid hazardous waste container within the designated area.
- Poly(I:C) is a biological reagent as well as a chemical; treated cultures containing live cells or organisms follow the laboratory's biological waste procedures for the biological component, but any culture that also contains a chemical hazard from this SOP is routed as chemical waste. Contact EH&S about mixed waste before generating it.
- Manage chemical and hazardous chemical waste separately from other waste streams such as biohazardous waste. Never autoclave chemical waste because it can produce hazardous chemical vapors, aerosols, and explosive reactions.
- The University of Washington Environmental Health & Safety Department has full responsibility for collection of hazardous waste for the University, all its campuses, and off-site locations; University laboratories cannot contract with an outside vendor to collect hazardous waste.
- To request a collection of chemical waste, submit a form on the Chemical Waste Disposal webpage on the EH&S website or directly in MyChem inventory. Contact EH&S at 206-616-5835 or chmwaste@uw.edu with questions.
- The laboratory holds two containers of resazurin sodium salt (15 g and 50 g) and 100 g of 4-methoxyphenol - quantities well beyond likely near-term use. The PI should review these annually and surplus the excess through EH&S rather than storing it indefinitely.
- Visit the Hazardous Material Disposal and Recycling webpage on the EH&S website for information on disposing, recycling and surplusing materials. Update the MyChem inventory when containers are emptied, received, transferred or surplusd.

Section 7 - Protocol

- Procedure A - Weighing solids and preparing stock solutions. Inside the certified fume hood, with the balance in the hood and an absorbent liner in place, wearing goggles, gloves as specified in Section 3, lab coat and sleeves: weigh the solid into a disposable weigh boat with a disposable spatula, transfer to a labeled tube, add the specified solvent (nuclease-free water, buffer, filtered seawater, or DMSO per SOP-02), cap, and vortex to dissolve. Aliquot into single-use tubes. Label each with compound, concentration, solvent, date and initials. Discard the weigh boat, spatula, tips and liner into the closed solid waste container. Wipe the balance and hood.
- Procedure B - Serotonin-induced spawning of bivalves. Prepare serotonin creatinine sulfate stock per Procedure A. Apply or inject the specified dose per the experimental protocol, wearing nitrile gloves, safety glasses and a lab coat. Contain all spawning seawater; collect it as chemical waste. Do not discharge to a drain or seawater return.
- Procedure C - Resazurin (AlamarBlue) viability assay. Prepare stock per Procedure A, protected from light. Add working solution to assay plates in the biological safety cabinet or at the bench, incubate, and read fluorescence. Collect spent plates and solutions as chemical waste.
- Procedure D - Poly(I:C) reconstitution and use. Reconstitute in sterile endotoxin-free saline or medium in the biological safety cabinet using aseptic technique, wearing nitrile gloves, safety glasses and a lab coat. Aliquot and freeze. Handle as a biologically active immune stimulant; avoid needlestick.
- Procedure E - 4-Methoxyphenol use. Weigh and dissolve per Procedure A inside the fume hood. Contain all solutions rigorously; assume that any release reaching water is an environmental release requiring EH&S notification.
- Prohibited: weighing any of these solids outside the fume hood; drain or seawater discharge of 4-methoxyphenol; handling of 4-methoxyphenol by a previously sensitized individual; use by untrained or unauthorized personnel.
- Refer to Section 2 of the UW Laboratory Safety Manual on the EH&S website for additional guidance on chemical management and preparation for use for particularly hazardous substances (PHSs).
- NOTE: Any deviation from this SOP requires approval from the Principal Investigator.

Section 8 - Special Precautions for Animal Use

Yes. Serotonin creatinine sulfate is used to induce spawning in bivalves, acetylcholine chloride may be used in physiological assays, and poly(I:C) may be used as an immune stimulant in live animals. All such use in live animals will be documented and approved by IACUC before the work begins, and the approved protocol number recorded here. Personnel handling dosed animals, their holding water and their waste must wear nitrile gloves, safety glasses and a lab coat; avoid aerosolizing holding water; keep dosing containers covered and posted; and collect all holding water, and any bedding, carcasses or cleaning materials from dosed animals, as chemical waste rather than discharging to a drain or seawater return. Tanks, nets and tubing used with dosed animals are labeled, dedicated where practical, and decontaminated per Section 10 before other use. 4-Methoxyphenol is not administered to animals.

IACUC protocol number (if applicable): _____

PARTICULARLY HAZARDOUS SUBSTANCE (PHS): YES - Sections 9, 10 and 11 are MANDATORY and are completed below.

Section 9 - Approvals Required

- All staff working with the compounds covered by this SOP must be trained on this SOP prior to starting work. They must also review the SDS for each compound, and the SDSs must be readily available in the laboratory (MyChem). All training must be documented and maintained by the PI or their designee.

- 4-Methoxyphenol and serotonin creatinine sulfate are particularly hazardous substances (reproductive toxicity Category 2). Written authorization from the Principal Investigator is required before any individual may work with them, and that authorization is recorded on the training roster in Section 12 with the PI's countersignature.
- Prerequisite training, all documented before independent work: (1) EH&S Laboratory Safety Training, initial and current refresher; (2) EH&S Managing Laboratory Chemical Waste training; (3) documented review of this SOP and the SDSs; (4) documented instruction on powder weighing technique inside a fume hood and on the aquatic-release prohibition for 4-methoxyphenol.
- A worker must complete at least one supervised run of Procedure A, observed and signed off by the PI or a designated experienced lab member, before weighing these solids independently.
- Live-animal use of serotonin, acetylcholine or poly(I:C) requires a current IACUC-approved protocol and PI authorization before any dosing occurs.
- Personnel who are pregnant, planning a pregnancy, or breastfeeding should consult the EH&S Reproductive Health in the Workplace resources and their healthcare provider and are encouraged to discuss alternative task assignment with the PI, because 4-methoxyphenol and serotonin creatinine sulfate are suspected reproductive toxicants. Such consultation is voluntary and confidential.
- Any individual who has experienced a skin reaction to 4-methoxyphenol must be reassigned away from that task.
- No medical examination is required for the work described because respirators are not used.

Section 10 - Decontamination

- Fume hood and balance: remove and bag the absorbent liner as hazardous solid waste; wipe the work surface, sash interior, balance pan and balance housing with a damp wipe (to capture dust rather than raise it), then detergent solution, then clean water, then 70% ethanol; dry.
- Equipment: wipe pipettors, racks and plate readers with detergent solution then 70% ethanol. Weigh boats and spatulas are disposable and are discarded rather than reused.
- Glassware: rinse three times inside the fume hood, collecting the first rinse in the appropriate waste container, then wash with detergent and water and rinse three times with deionized water.
- Animal exposure equipment (tanks, nets, tubing), if used: drain into the chemical waste container, wash with detergent and water, and rinse three times with clean water before returning the equipment to general use.
- Controlled areas: there are no glove boxes or restricted-access hoods in FTR 209; the designated areas in Section 11 are the controlled areas and are decontaminated at the end of each session and monthly regardless of use.
- Cleaning solutions and materials: laboratory detergent (Alconox or Contrad) solution, clean water, 70% ethanol, lint-free wipes, disposable absorbent liners.

Section 11 - Designated Area

- DESIGNATED AREA 1 - weighing solids and preparing stock solutions: the certified chemical fume hood in FTR 209, with the analytical balance inside the hood and a disposable absorbent liner in place.
- DESIGNATED AREA 2 - dosing, assay setup and treated-sample handling: the biological safety cabinet and the immediately adjacent bench in FTR 209, over an absorbent liner.
- DESIGNATED AREA 3 - if live-animal exposures are conducted: the posted exposure tank area with secondary containment, restricted access, and labeled dedicated tanks, nets and tubing.
- Designated areas used for 4-methoxyphenol and serotonin creatinine sulfate are posted with a 'DESIGNATED AREA - PARTICULARLY HAZARDOUS SUBSTANCE - REPRODUCTIVE TOXICANT' sign listing the PI, contact number, required PPE and the authorization requirement. The 4-methoxyphenol sign adds 'TOXIC TO AQUATIC LIFE - NO DRAIN DISPOSAL'.

- Waste containers for these compounds are kept within the designated areas in secondary containment.
- Storage: labeled closed secondary containers in the toxics cabinet and the refrigerator/freezer in FTR 209, posted with the PHS designation where reproductive toxicants are stored.

Section 12 - Documentation of Training

- Prior to using substances included in this SOP, laboratory personnel must be trained on the hazards described in this SOP, how to protect themselves from the hazards, and emergency procedures.
- Ready access to this SOP and to a Safety Data Sheet for each hazardous material described in the SOP must be made available in the lab space(s) where these substances are used.
- The Principal Investigator, or Responsible Party if the activity does not involve a PI, must ensure that their laboratory personnel have attended appropriate laboratory safety training (and refresher training where applicable).
- Training must be repeated following any revision to the content of this SOP.
- Training must be documented. This training sheet is provided as one option; other forms of training documentation (including electronic) are acceptable but records must be accessible and immediately available upon request.
- Because this SOP covers a particularly hazardous substance, the Principal Investigator countersigns each entry below to record the written authorization required by Section 9.

I have read and understand the content of this SOP:

[illegible]